

Changes in momentum

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 2 starred (★) questions first.

	QUESTION	ANSWER	✓
1	State how the momentum of an object can be changed.	By a resultant force acting on it.	☐
2 ★	How is force related to momentum?	Force is the rate of change of momentum.	☐
3 ★	Explain how air bags, seat belts and crash mats reduce the force in a collision.	They increase the time taken for the momentum to change, which reduces the force.	☐
4	State the equation linking force, change in momentum and time.	force = change in momentum ÷ time taken ($F = m \Delta v \div \Delta t$)	☐